THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 6 — MATHS · NCERT-TYPE UNIT TEST SET 3

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. In the pattern of cube numbers $1, 8, 27, 64, \ldots$ the term after 125 is: 1

(a) 140 **(b)** 161 **(c)** 216 **(d)** 36

2. Points lying on the same line are called: 1

(a) collinear points **(b)** adjacent points **(c)** parallel points **(d)** vertex points

3. In a supercell pyramid each number is the sum of the two numbers just below it. The bottom row is 4, 6, 3. The top number is: 1

(a) 19 **(b)** 25 **(c)** 13 **(d)** 26

4. A pictograph where one symbol = 10 shows 5 full symbols and a half symbol. The number shown is: 1

(a) 50 **(b)** 60 **(c)** 51 **(d)** 55

5. The smallest prime number that is a factor of 105 is: 1

(a) 1 **(b)** 3 **(c)** 5 **(d)** 7

6. In the Virahanka (Fibonacci) sequence $1, 2, 3, 5, 8, \ldots$ the 8th term is: 1

(a) 55 **(b)** 34 **(c)** 21 **(d)** 47

7. What is the smaller angle between the hands of a clock at 6 o'clock? 1

(a) 150 **(b)** 180 **(c)** 270 **(d)** 210

8. Add 25 and its reverse 52. The sum is: 1

(a) 88 **(b)** 99 **(c)** 77 **(d)** 66

9. The number of times a value occurs in the data is its: 1

(a) key **(b)** range **(c)** scale **(d)** frequency

10. Which of the following is divisible by 8? 1

(a) 5128 **(b)** 5130 **(c)** 5129 **(d)** 5131

11. A complete graph has 4 points, each joined to every other point. How many lines are there? 1

(a) 6 **(b)** 16 **(c)** 12 **(d)** 10

12. A ray turns anticlockwise through 90° from the horizontal line to the right. The type of angle formed is: 1

(a) right **(b)** obtuse **(c)** acute **(d)** straight

13. Kaprekar step: for 8261, (largest arrangement) - (smallest arrangement) of its digits is: 1

(a) 7344 **(b)** 7353 **(c)** 7362 **(d)** 7443

14. In a pictograph one symbol stands for 20 units. How many symbols (counting a half symbol as half) are needed to show 70? 1

(a) 3 **(b)** 4 **(c)** 3.5 **(d)** 1400

15. Which pair is a pair of co-prime numbers? 1

(a) 16 and 24 **(b)** 14 and 25 **(c)** 9 and 15 **(d)** 14 and 21

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the sum $1 + 3 + 5 + \ldots$ of the first 13 odd numbers without adding one by one. 2

17. A line has 4 marked points A, B, C, ... Write how many different line segments can be named. 2

18. Write the hailstone (Collatz) sequence for 6. 2

19. Write the frequency of each number in the data: 1, 2, 4, 2, 2, 3, 3, 2, 4, 1, 4, 2, 1, 4. 2

20. Are 14 and 15 co-prime? Give the reason. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the first 4 common multiples of 4 and 6. 3

22. Add the triangular numbers 55 and 66. What do you notice about the answer? 3

23. A student turns clockwise from facing North through 180°. In which direction (N, E, S, W or in between) is she now facing? Name the type of the angle of turn. 3

24. Show that the Kaprekar routine on 4821 reaches 6174. 3

OR

Study the table and answer: 3

(i) the largest,
(ii) the smallest,
(iii) the total.

Item	Number
Orange	65
Apple	60
Mango	45

25. Use the divisibility test for 11 to check whether 5412 is divisible by 11. 3

OR

Write the sum $1^3 + 2^3 + \cdots + 4^3$ and show it equals the square of $1 + 2 + \cdots + 4$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Data of 14 students' number of siblings: 5, 3, 5, 5, 5, 1, 2, 2, 2, 4, 0, 3, 3, 4. Prepare a frequency table, and find the most common number of siblings and how many students have more than 2 siblings. 5

OR

A number is divisible by both 3 and 4. Show that it is divisible by 12 when 3 and 4 are co-prime. Give an example with 60. 5

27. A group of 7 friends shake hands with each other exactly once. Using the idea of a complete graph, find the number of handshakes. Show the working. 5

OR

A ray OA makes 45° with OB and another ray OC makes 135° with OB on the same side of OB, with OC beyond OA. Find $\angle AOC$, and name the types of $\angle AOB$, $\angle BOC$ and $\angle AOC$. 5

END OF QUESTION PAPER **Total: 50 Marks**

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